

EMEM: A Content-Addressed, Verifiable Earth-Memory Protocol for AI Agents over Foundation-Model Embeddings

Agents hand each other evidence references, not paraphrases.

The receiver resolves the reference, re-hashes the record, re-hashes the source. A signature fixes the record's bytes, not the measurement.

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Agentic AI for Earth Observation · BIFOLD and ESA Φ-lab · Berlin · Poster Session 1 · 19 Oct 2026 · emem.dev · github.com/Vortex-AI/emem (Apache-2.0)

1 What survives an agent handoff?

Handed prose, agents acted on 254 of 276 corruptions; handed a reference they check, on 0 of 300.

A relay or a faulty signer corrupts one signed NDVI record in one of 16 ways; Agent B resolves, re-hashes, binds, checks, recomputes and re-reads.

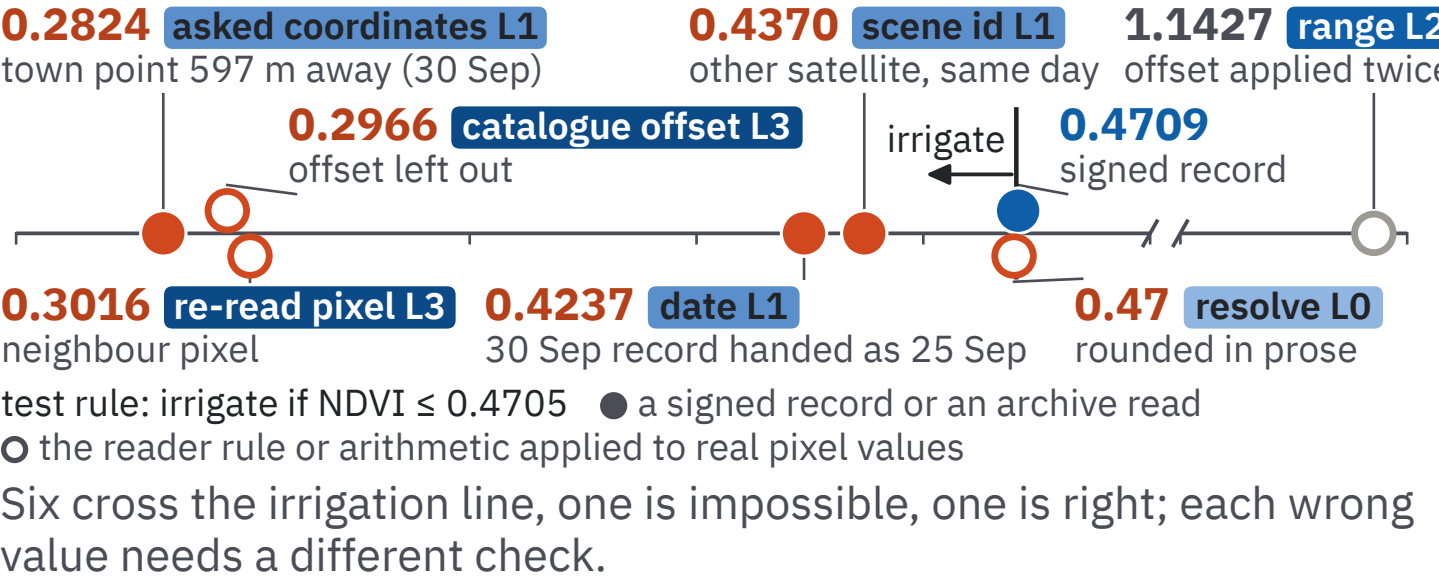


With a model as B and a forged cell, Claude Haiku 4.5 declined 5 of 5; Qwen2.5-3B dropped the cell from all 24 calls and acted 5 of 5: a reference protects only a receiver that keeps it whole and obeys a refusal.

2 What can one NDVI become between agents?

Eight answers; one is right

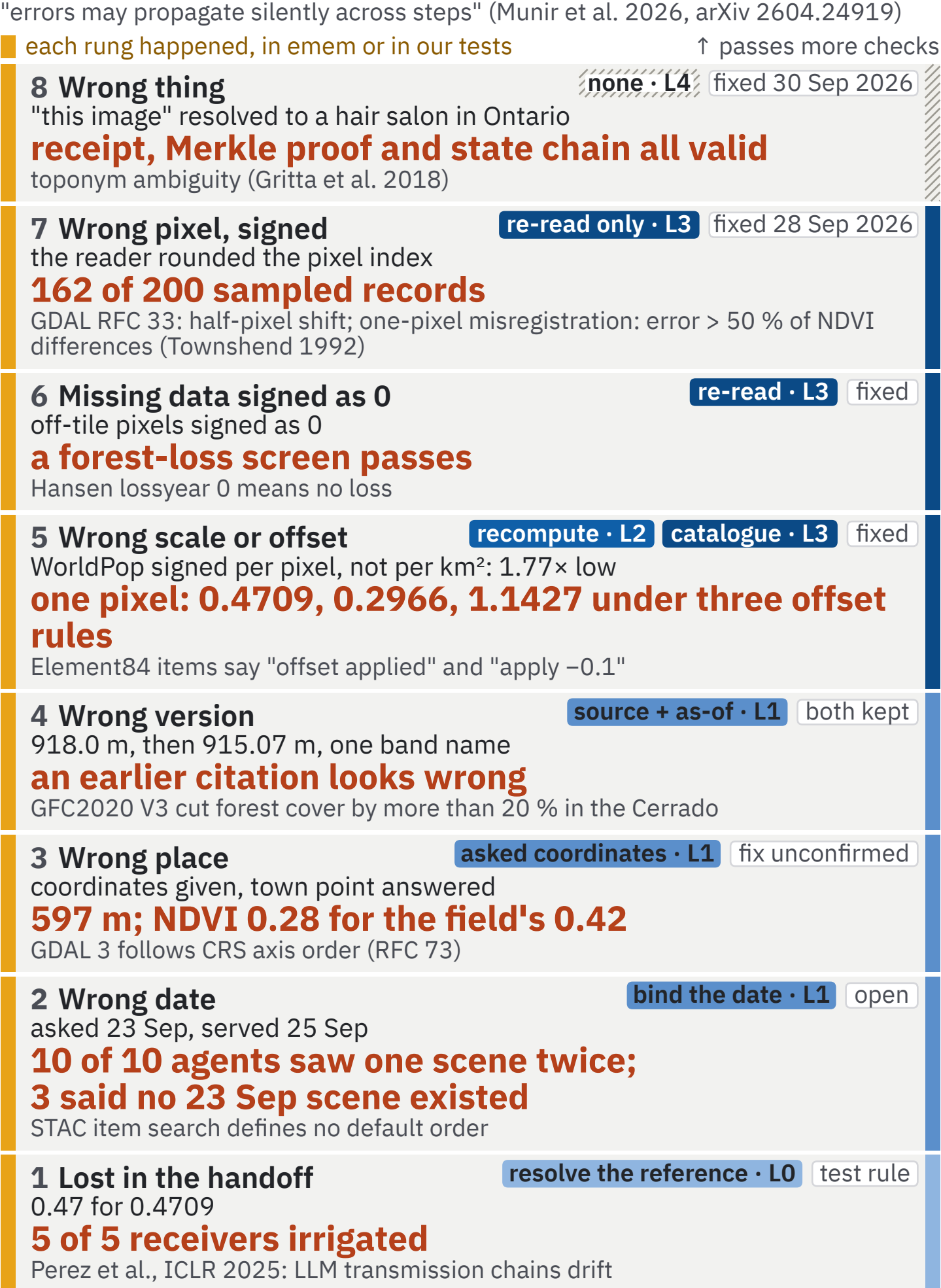
One question, NDVI at the Keylong field on 25 Sep 2026; real records, archive pixels and handling rules give eight values.



3 Why does an agent handoff need a record?

Our own errors name the checks

Each error happened inside emem, a system built for evidence, or in our tests; each has a published precedent.



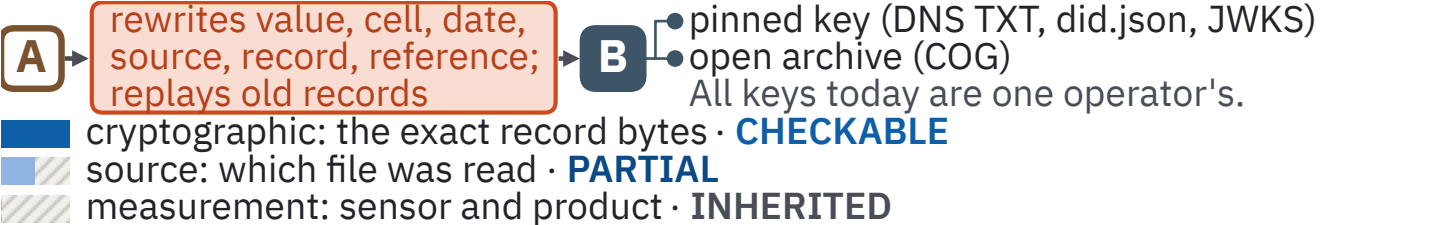
Signatures prevented none; the record made each checkable at a named layer, except the top row. A check protects only if it runs: emem's resolver once reported a cell match it never tested; one framework adapter returns a refusal as plain text; MAST: no or incomplete verification (FM-3.2).

We ask four questions

- RQ1** Can a receiver detect corruption without trusting the sender? → 1, 5
 - RQ2** Which corruption classes can it detect? → 5
 - RQ3** Does a reference beat prose, JSON, retrieval or an opaque id? → 1, 5
 - RQ4** What stays unverifiable with the reference intact? → 6, 7
- H1** A checked reference refutes corruptions that keep the stated value.
H2 Only a source re-read catches a signer's wrong pixel. **H3** A historical reference keeps the cited state.

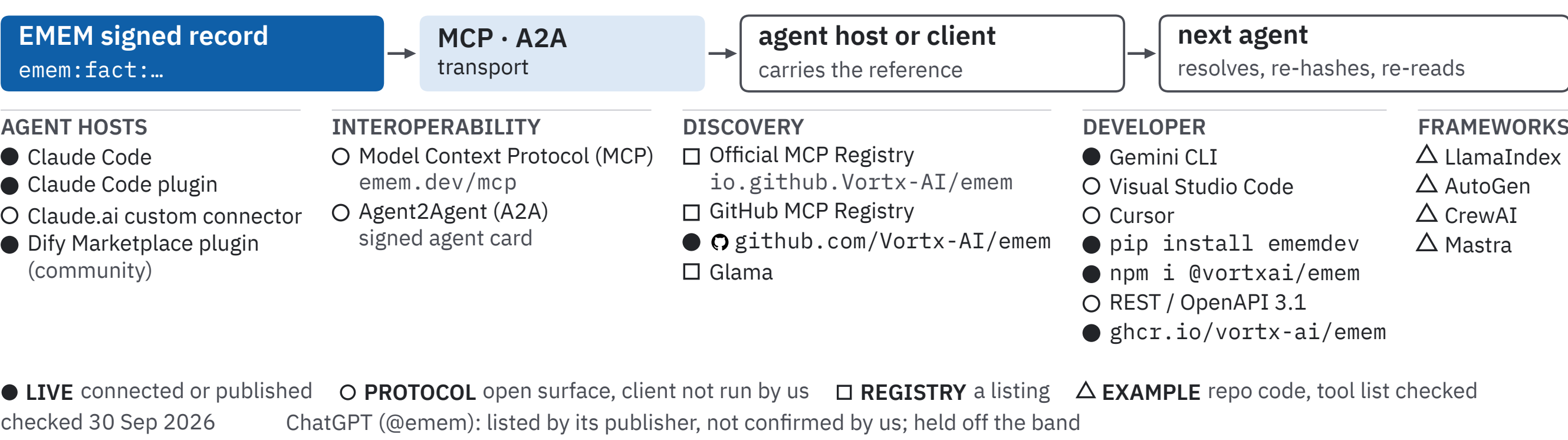
A relay can rewrite, not sign

A relay can rewrite what it carries but cannot sign under the pinned key or match an address; a signer's misread pixel is caught only by a re-read. Out of scope: a compromised key, a wrong sensor, the entity meant, the decision.

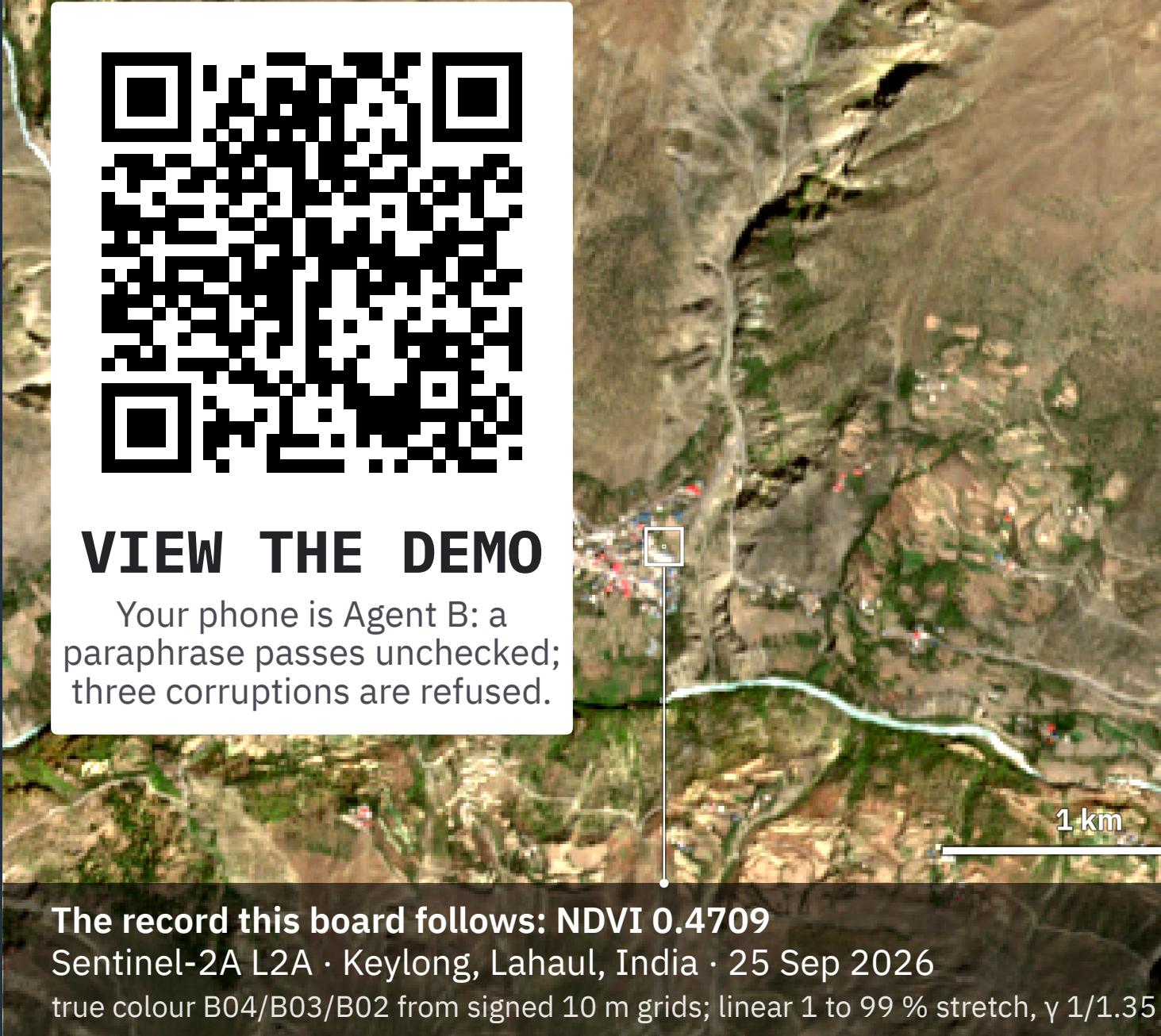


12 One evidence protocol, multiple agent runtimes.

One reference resolved to one address and one value through 11 client paths, with receipt signatures checked on 9 (30 Sep 2026). A listing is not an integration. Not run by us: ChatGPT, claude.ai, Dify, VS Code, Cursor.



Mechanisms read from emem's code; where its docs differ, the code is printed. emem.dev at commit 8e9b401; measurements 29 Sep to 1 Oct 2026. Extending verification to execution: reference harness, no spacecraft enrolled. · Same pattern in software supply chains: Sigstore, RFC 9162, SCITT (RFC 9943). Closest agent memory: ARC (arXiv 2607.25066). · Sentinel-2 Products Specification (ESA) · Copernicus DEM GLO-30/90 · Hansen et al. 2013, GFC v1.13 · JRC GFC2020 V3/V4, TMF · ESA CCI Biomass v7 · Reg. (EU) 2023/1115 · BLAKE3 · RFC 8032 · RFC 6962/9162 · STAC 1.1 · openEO 1.3 · W3C PROV-DM · C2PA 2.4 · MCP 2025-11-25 · A2A 1.0 · Perez et al., ICLR 2025 · Munir et al. 2026 (2604.24919) · Cemri et al. 2025, MAST (2503.13657) · Dang et al. 2026, ARC (2607.25066) · Townshend et al. 1992 · GeoGuard (NASA-IMPACT) · Prithvi-EO-2.0 · TESSERA (2506.20380)



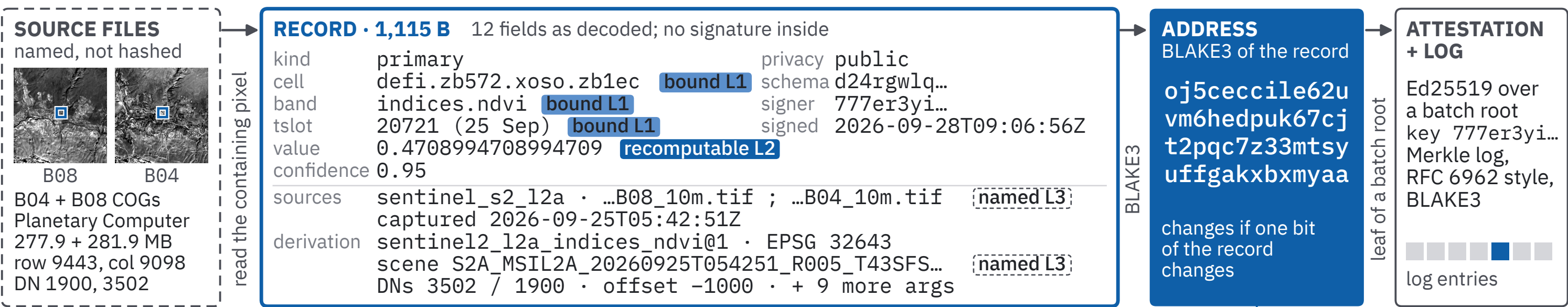
The record this board follows: NDVI 0.4709
Sentinel-2A L2A · Keylong, Lahaul, India · 25 Sep 2026
true colour B04/B03/B02 from signed 10 m grids; linear 1 to 99 % stretch, y 1/1.35

4 What exactly is handed over?

The address names the record, not the pixel.

The receiver gets 84 characters; everything else it fetches and checks.

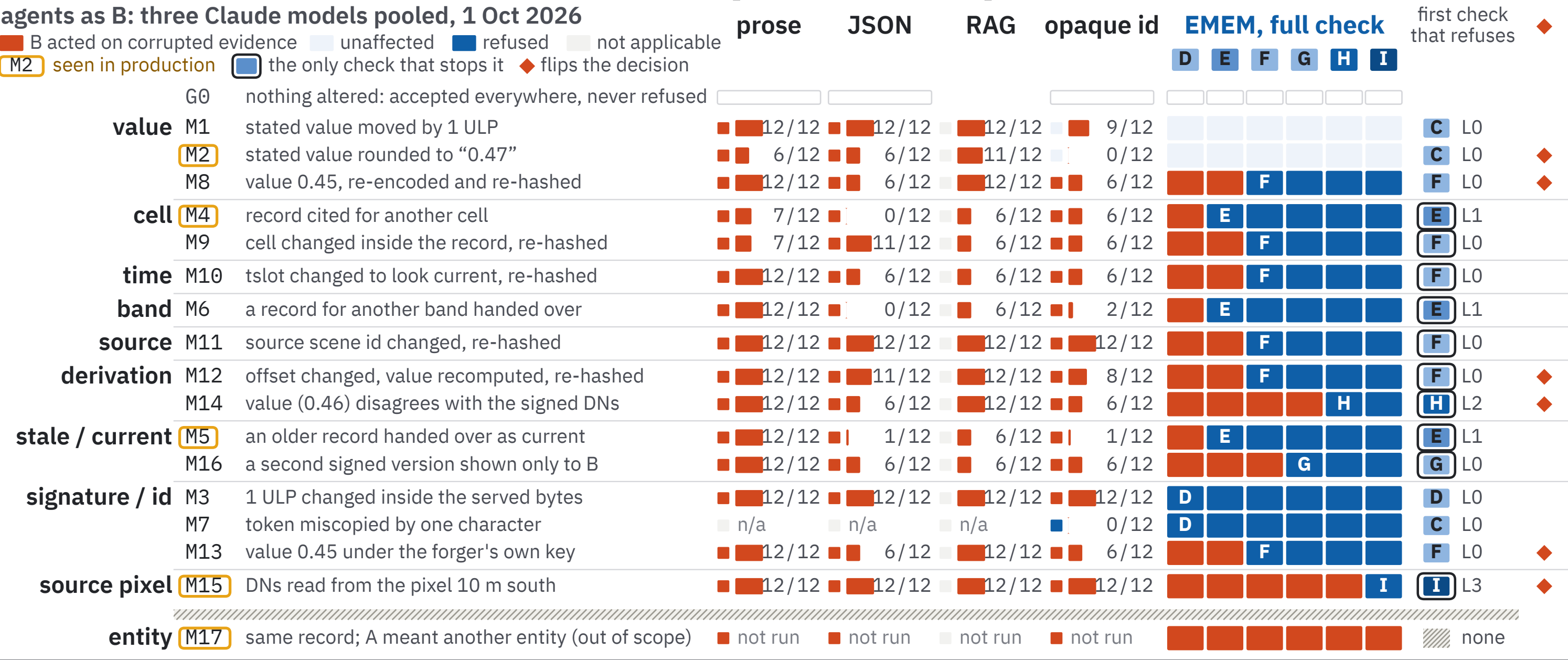
INSPECT THE RECORD The 1,115 bytes behind the main example, decoded.



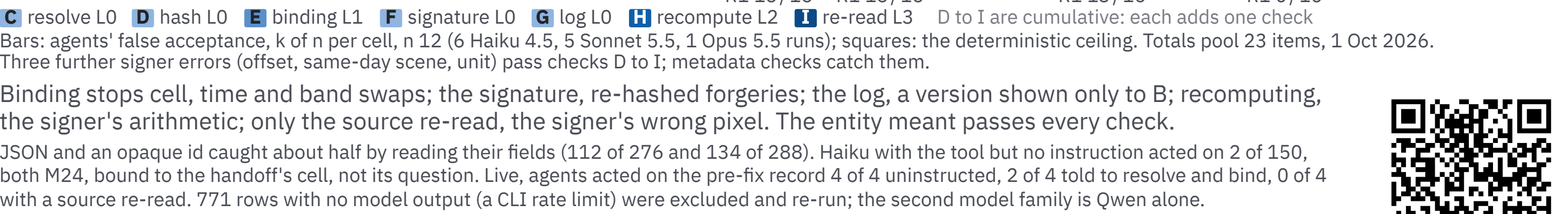
The address is the BLAKE3 hash of the 1,115-byte record, which names its source files and the point it read but holds no hash of them; a batch signature covers such addresses. Re-signing makes a new record: one Bengaluru value has seven.

5 Which check stops which corruption?

Each of five checks alone stops a corruption.



B acts on corrupted evidence

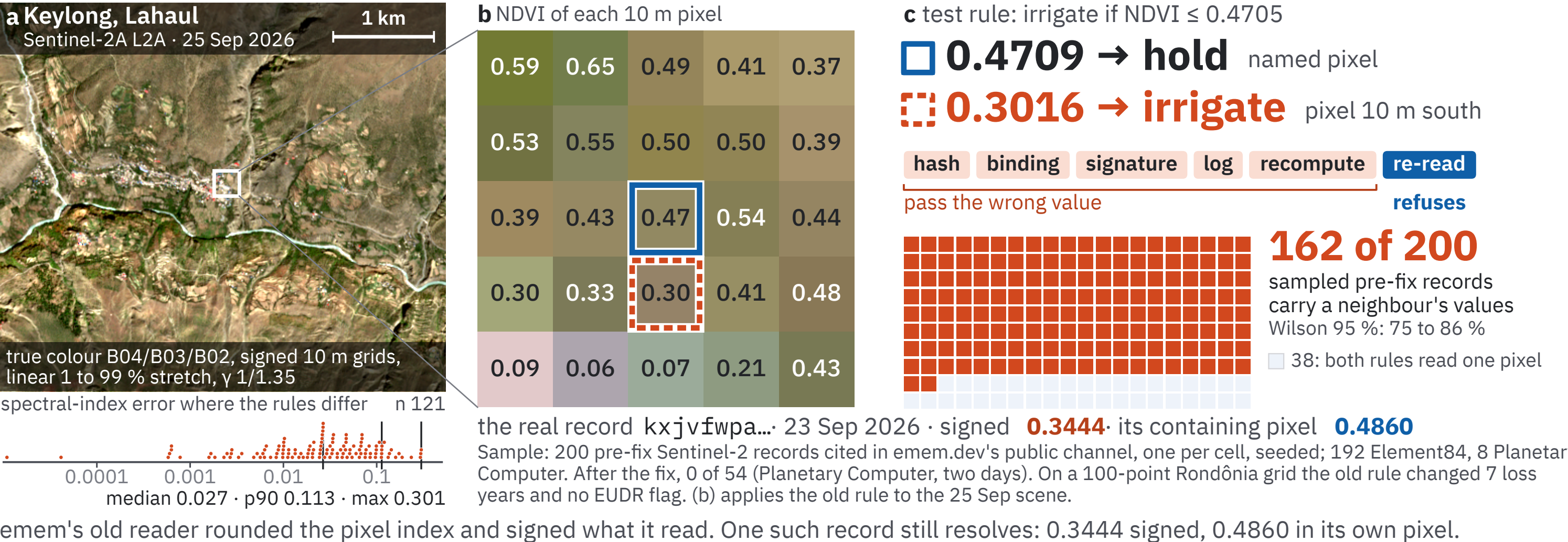


RE-RUN THE TEST Data and scripts behind this board's numbers, with one re-run command.

6 Can a valid signature carry a wrong measurement?

The right record, the wrong pixel.

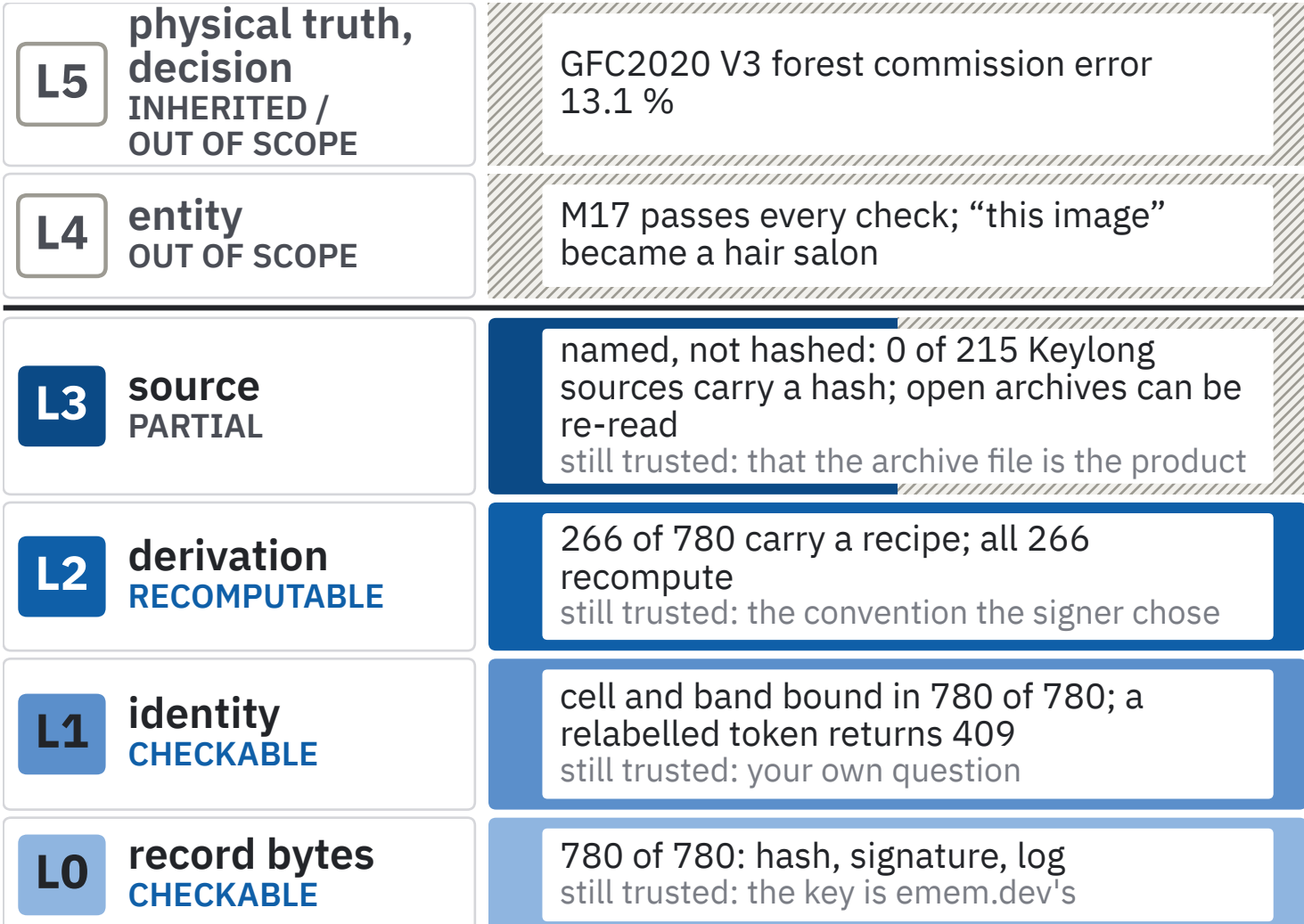
A valid signature can faithfully preserve a wrong measurement when the reader itself selected the wrong source pixel.



7 What does a checked reference establish?

Checks stop at the source

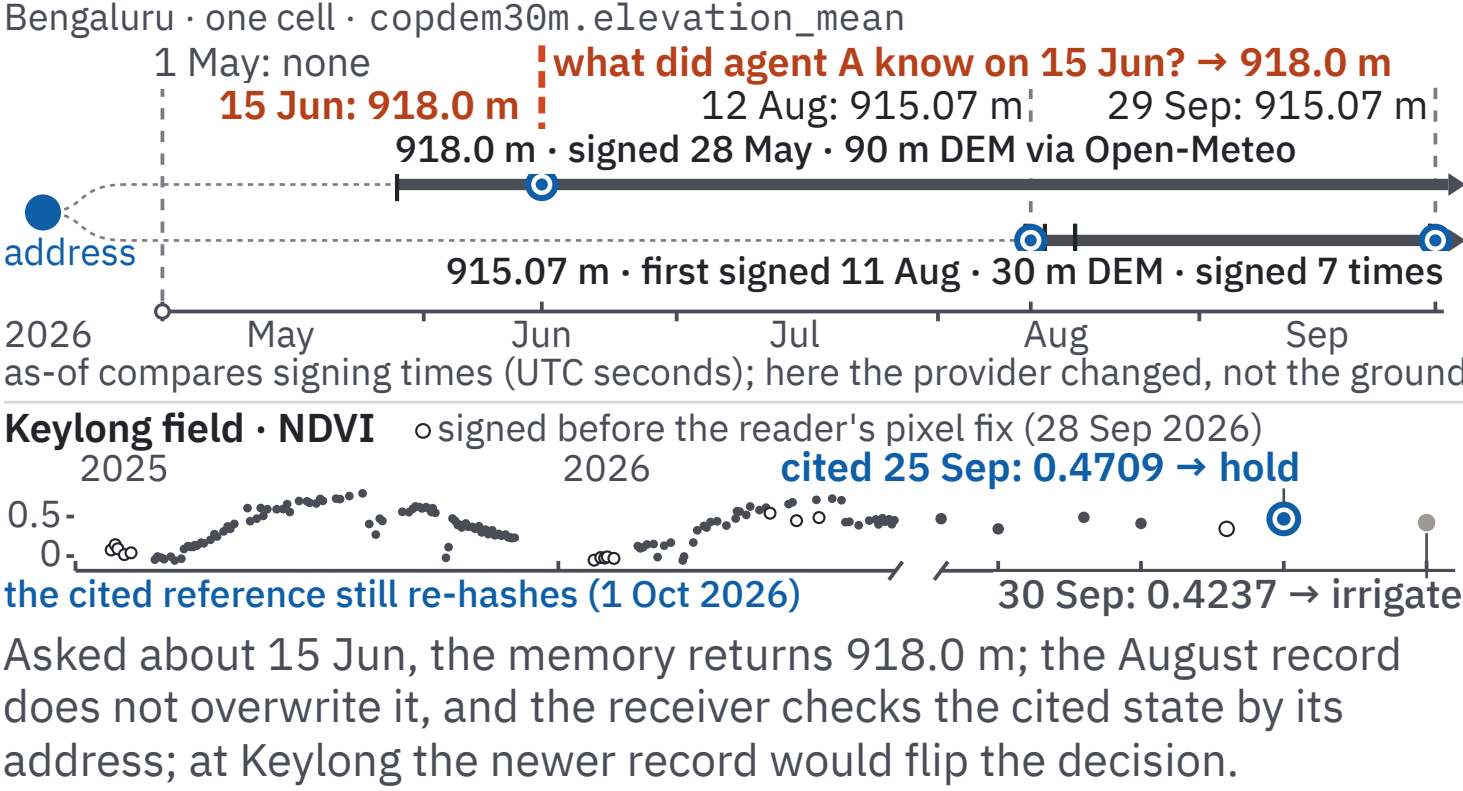
Each rung names what it still trusts.



Of 780 records pulled on 30 Sep 2026, 780 pass L0 and L1; the 266 with a recipe all recompute at L2; their source files are named, not hashed. No reference establishes the entity, the sensor's accuracy or the decision.

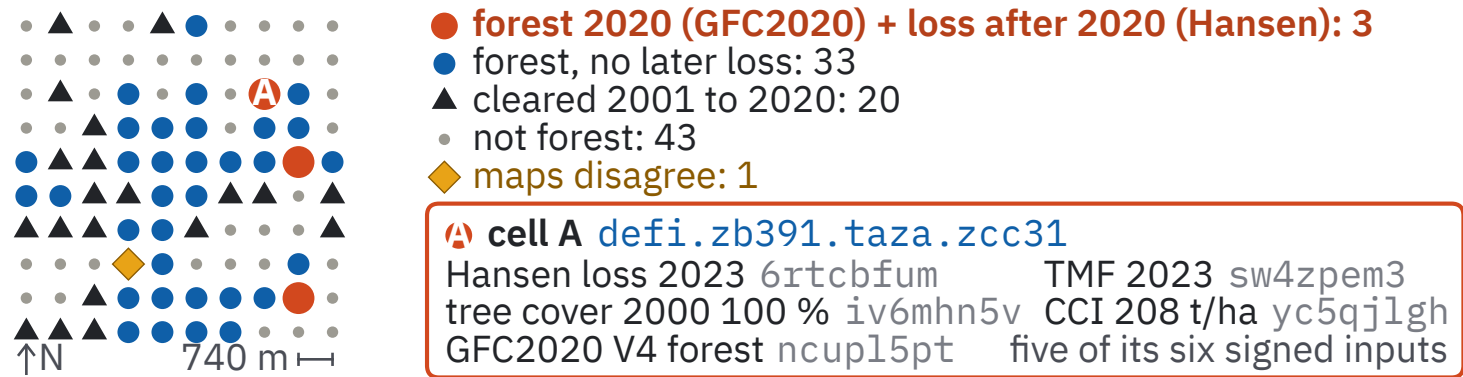
8 What did the agent know on 15 Jun?

Memory keeps what was cited



9 Can a screen be re-checked input by input?

An auditor re-runs each input



Point samples, not parcel polygons; not a regulatory determination.

100 point samples 740 m apart, 600 signed records, each re-hashed; TMF agrees on 1 of 3 flags.

10 Where are the foundation-model embeddings?

Vectors are records too

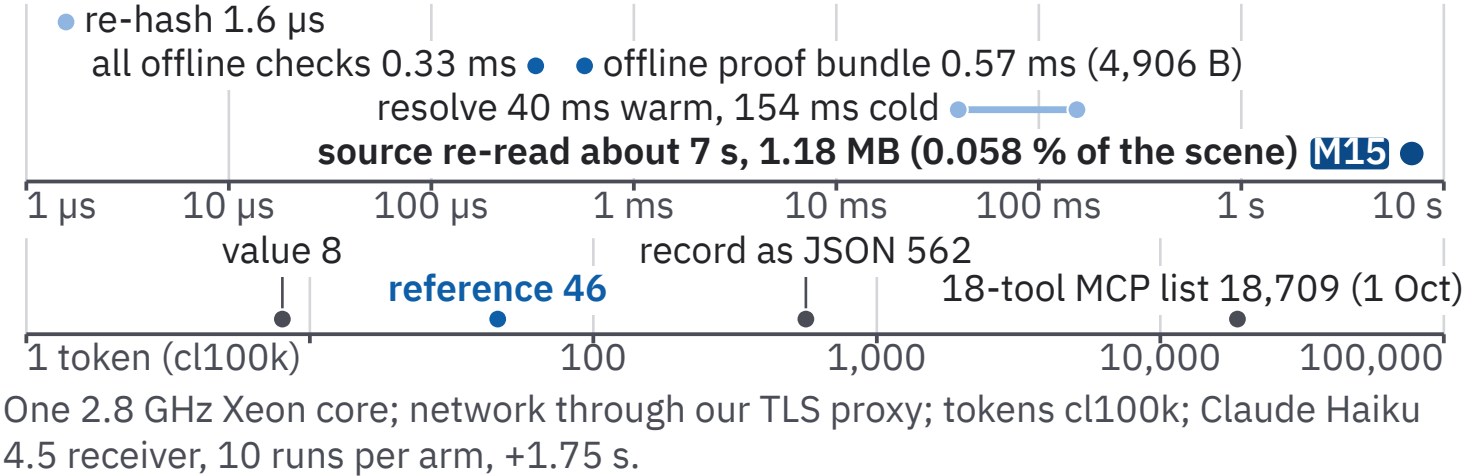
Derived representations are addressable too: this cell's Prithvi-EO-2.0 vector names its checkpoint digest inside the hashed record; the TESSERA vector, only a path and year. Both still resolve; emem.dev lists its encoders as retired.



11 What does checking cost?

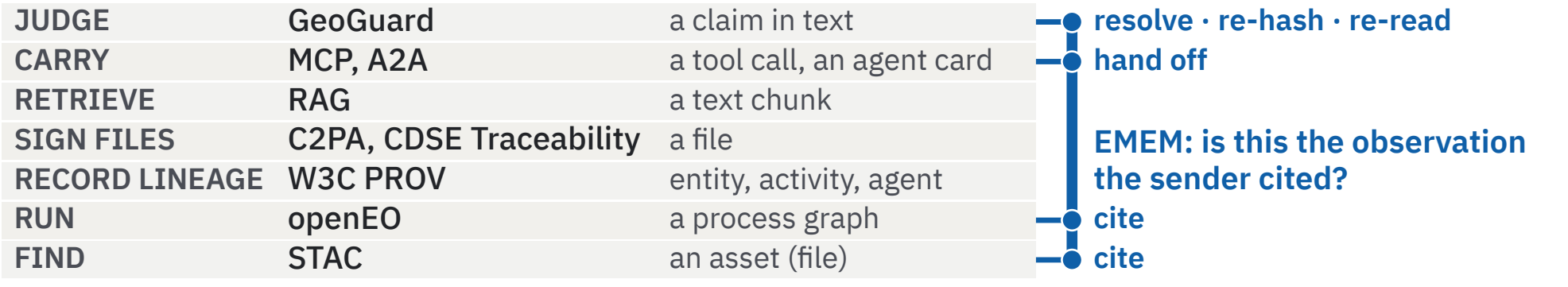
The source re-read costs most

A record check takes 0.33 ms of CPU; a pixel re-read moves 1.18 MB in about 7 s, and only it refutes the wrong pixel. A reference costs 46 tokens, the value 8; a model receiver read 2.1x the tokens prose costs, same decisions.



13 EMEM relies on these layers

A STAC item identifies a file; EMEM identifies the observation an agent cited, for the next agent to re-check. GeoGuard judges the claim; EMEM fixes the evidence it cites.



A signed observation reference left the receiver acting on none of the 16 corruptions in our suite, and agents with a checked reference acted on 0 of 300; entity, sensor and decision stay beyond its checks.

READ THE METHODS vortex-ai.github.io/esa_poster

Definitions, denominators, models, and what our tools generated.

